MATH 11 — ELEMENTARY STATISTICS

§1.5

Statistical Studies

Where data comes from, and what it lets us claim

Today's Goal

- Distinguish between observational studies and experiments
- Identify different types of observational studies
- Identify the key components of an experiment
- Understand the importance of randomization and blinding in experiments

What You Will Be Graded On

LT.4 Statistical Studies

I understand what experiments and observational studies are. I can use this understanding to provide detailed descriptions of examples of each.

A Headline You Have Seen

“People who eat chocolate live longer.” Researchers asked thousands of adults what they ate, then followed who was still alive years later.

Does chocolate make people live longer?

Nobody was told to eat chocolate. People who buy it may also have more money for good food, housing, and health care.

The study shows a **link**, not a **cause**.

Watching, Without Stepping In

Definition 1.5.1

An **observational study** is a study in which researchers observe and collect data without intervening or manipulating any variables.

The chocolate study is one of these. The researchers changed nothing.

Changing Something on Purpose

Definition 1.5.2

An **experiment** is a carefully designed study in which researchers deliberately manipulate one or more variables to observe the effect on other variables.

A chocolate experiment would *assign* some adults to eat chocolate and others not to.

Ask One Question

The key distinction is whether the researchers *intervene*:

- In an **observational study**, researchers only observe what happens naturally.
- In an **experiment**, researchers actively change something and measure the response.

Why the Difference Matters

Warning 1.5.3

Only experiments can establish **cause-and-effect** relationships. Observational studies can show associations or correlations, but they cannot prove that one thing causes another.

Naming the Study Type

Example 1(a).

A campus dietitian records what each student picks in the dining hall for one semester, and tracks how often those students report feeling tired.

Experiment or observational study?

She serves nobody a different meal. **Observational study.**

Naming the Study Type

Example 1(b).

A sleep researcher randomly assigns 60 volunteers to sleep either 6 hours or 8 hours a night for two weeks, then measures each one's reaction time.

Experiment or observational study?

He decides who sleeps how long, then measures the response. **Experiment.**

Now You Try: Which Kind Is It?

You Try 1.5.1. Name the study type in each scenario.

1. Blood pressure is compared for coffee drinkers and non-drinkers. **Observational.**
2. Volunteers are randomly given a standing desk or a regular desk for a month. **Experiment.**
3. A school compares each student's social media hours with their grades. **Observational.**

A Snapshot of Right Now

Definition 1.5.5

A **cross-sectional study** is an observational study in which data are collected at one point in time (or over a short period).

Example: a poll taken this morning about how people rate city bus service.

Looking Back

Definition 1.5.7

A **retrospective study** (also called a *case-control study*) is an observational study in which data are collected from the past, often by examining existing records or interviewing participants about past events.

Example: reading the old training logs of runners who later broke a bone.

Looking Forward

Definition 1.5.9

A **prospective study** is an observational study in which participants are followed forward in time, with data collected as events occur in the future.

Nothing has happened yet when the study starts. You wait and record.

Same People, Again and Again

Definition 1.5.10

A **longitudinal study** is an observational study in which data are collected from the same subjects repeatedly over an extended period of time.

One Group, Followed

Definition 1.5.11

A **cohort study** is a type of longitudinal study in which a group of individuals (a cohort) sharing a common characteristic is followed over time to observe specific outcomes.

What Each Word Describes

Warning 1.5.12

These three study types describe different (but overlapping) aspects of a study:

- **Prospective** describes the *temporal direction* — data is collected going forward in time.
- **Longitudinal** describes the *measurement pattern* — the same subjects are measured repeatedly.
- **Cohort** describes the *subject selection* — a group sharing a common characteristic is followed.

Can One Study Be All Three?

A team enrolls 500 newly hired nurses and measures each one's burnout score every year for five years.

Which of the three words fit this study?

All three. Forward in time, the same nurses each year, one shared trait.

The Two Variables

- **Explanatory Variable** (also called *independent variable*): The variable that researchers manipulate or change.
- **Response Variable** (also called *dependent variable*): The variable that researchers measure to observe the effect of the explanatory variable.

You change the explanatory variable. You measure the response variable.

Treatments and Groups

- A **treatment** is the specific condition or value of the explanatory variable applied to subjects.
- **Experimental Group**: The group of subjects that receives the treatment being tested.
- **Control Group**: The group of subjects that does not receive the treatment, serving as a baseline for comparison.

Three Rules for a Good Experiment

1. **Replication:** Test the treatment on many subjects, not one. One patient who feels better proves nothing.
2. **Randomization:** Assign subjects to groups at random, so group differences come from the treatment.
3. **Blinding:** Keep participants, and sometimes researchers, from knowing which treatment they get.

Why Hand Anyone a Sugar Pill?

In a drug study, one group gets the real drug. The other gets a pill with no medicine in it.

Why bother giving that group a pill at all?

People often feel better just from believing they are treated. Give both groups a pill and the belief matches, so only the medicine differs.

Believing You Are Treated

Definition 1.5.14

The **placebo effect** occurs when subjects experience a perceived improvement in symptoms simply because they believe they are receiving treatment, even if the treatment has no actual therapeutic effect.

A **placebo** is an inactive treatment, like a sugar pill, that looks identical to the real one.

Who Is Kept in the Dark?

- **Single-Blind**: Participants do not know which treatment they receive, but researchers do.
- **Double-Blind**: Neither participants nor the researchers directly working with them know which treatment is administered.
- **Triple-Blind**: Participants, researchers, and data analysts are all unaware of treatment assignments.
- **Investigator-Blind**: Participants know which treatment they receive, but researchers do not. This is used when blinding participants is impossible (e.g., surgical vs. non-surgical treatment).

Taking an Experiment Apart

Example 2.

A team tests whether a new tutoring app lowers test anxiety. They recruit 180 students and randomly split them into two groups of 90: one uses the new app, the other uses a look-alike app with the tutoring turned off. Students are not told which app they have, and the counselor who scores the survey is not told either. Anxiety is measured before and after 8 weeks.

Name the six components.

The Six Components

- **Explanatory variable:** which app a student uses (real tutoring app or look-alike)
- **Response variable:** test anxiety score
- **Treatments:** the new tutoring app and the look-alike app
- **Control group:** the 90 students using the look-alike app
- **Experimental group:** the 90 students using the new tutoring app
- **Blinding:** double-blind — neither the students nor the counselor scoring the survey knows which app a student had

Now You Try: Name the Parts

You Try 1.5.2. A gym tests whether a new warm-up routine reduces injuries. 240 members are randomly split into two groups of 120: one does the new warm-up, the other does the usual one. Members are not told which routine is the new one. Injuries are counted for six months.

Name all six components.

You Try 1.5.2: Checking Your Answers

- **Explanatory variable:** which warm-up routine a member does
- **Response variable:** number of injuries in six months
- **Treatments:** the new warm-up routine and the usual warm-up
- **Control group:** the 120 members doing the usual warm-up
- **Experimental group:** the 120 members doing the new warm-up
- **Blinding:** single-blind — members do not know, trainers do